



A novel wind turbine data imputation method with multiple optimizations based on GANs



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ABSTRACT

In the rising research and applications of data-driven technologies in mechanical systems, data missing has always been a serious problem. The problem of data missing on a large scope has brought grave challenges to the operation and maintenance of the machineries, such as wind turbines (WTs). In this paper, a WT data imputation method with multiple optimizations based on generative adversarial networks (GANs) is proposed. First, to tackle the problem of data missing in large-scale WTs, a conditional GANs-based deep learning generative model is designed according to data features. Second, the permutation of the training data is optimized, so that the convolutional kernel can be better applied. The optimization problem is creatively transformed to a travelling salesman problem (TSP), and two optimization functions are proposed based on data features. Then, the relationship between the training data and the convolutional kernel is studied, and two restrictions are put forward to make the imputation model more effective. Finally, four data imputation experiments and two optimization experiments are carried out using real WT data. The experiment results verify the effectiveness of the proposed method.

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1. Introduction

As a complex mechanical system, wind turbine (WT) has gained a rapid development in recent years. In the studies of WTs and the related mechanical systems, many subjects [1–5] (including condition monitoring, fault diagnosis, fault prognosis, prediction of remaining useful life (RUL), etc.) play an important role in the daily operation and maintenance (O&M) of machineries. Most of these subjects require complete data. However, due to the harsh environment where some mechanical systems (especially for WTs) are installed, data missing problem occurs frequently [6]. Also, high quality data is especially required in the recently widely-studied methods based on artificial intelligence (AI) [7–9]. Therefore, the imputation of missing data has become a very important research topic in WT mechanical system studies.

Complete data are essential for many methods in mechanical applications. In the field of fault detection, many effective methods, such as spectrum kurtosis (SK) [10], local mean decomposition (LMD)[11], intrinsic time scale decomposition (ITD) [12], variational mode decomposition (VMD) [13] and singular spectrum decomposition (SSD) [14], have achieved good results in detecting many types of mechanical faults [15–18]. Moreover, the complete data are also necessary for many methods in the field of RUL prediction of machine components [19]. Therefore, data imputation is significant for mechanical applications.

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The data collected from WT's basically include high frequency data [19] (such as vibration data) and low frequency data [20] (such as supervisory control and data acquisition data, SCADA data for short). The SCADA data usually contain hundreds of variables at a long interval (10 min for example). The methods using SCADA data have been widely studied [21–24]. Due to various data types and the long collection interval, the missing SCADA data could cause heavy information loss, which would lead to more serious problems. For these reasons, this paper aims to impute the WT SCADA data.

Data imputation methods in industrial fields mainly consist of two categories: methods based on statistics and methods based on data mining [25].

In statistics-based methods [26–30], statistical features and the probability of missing data are studied to impute the missing data. In [26], an empirical framework based on autoregressive integrated moving average (ARIMA) model was presented and the wind speed model was calibrated. In [27], a support vector machine enhanced Markov model was developed for short-term wind power prediction. In [28], a comprehensive and statistical framework in a model free setting for a complete treatment of localized data corruptions was proposed. In [30], a spatial-temporal method based on multiple linear regression model was put forward to impute the traffic speed of the segments without sensor data. In [31], a statistical imputation method was proposed to reconstruct air quality datasets.

In data mining-based methods [32–36], data-driven technologies and artificial intelligence (AI)-based methods are used to impute the missing data. In [32], a transfer-based additive least squares support vector machine (LS-SVM) was proposed to clean the training data and improve data quality. A hybrid intelligent method for short-term wind power forecast and uncertainty analysis was presented based on support vector regression (SVR) in [33]. In [34], a algorithm called FINNIM that iteratively interpolates missing values by utilizing complete and incomplete instances was put forward. In [35], a method of combining a decision tree algorithm and an expectation-maximization algorithm was proposed to impute missing values. In [37], an imputation method based on two data-driven algorithms was presented. In recent years, the study of deep learning methods has gained a rapid growth, and many data imputation methods have been developed accordingly. In [38,39], convolutional neural network (CNN) has been adopted to impute the missing data. In [40,41], the prediction and identification methods based on deep belief network (DBN) and denoising autoencoder (DAE) were proposed.

However, the data imputation effect using these methods would decline in some complex conditions. There are many challenges in imputing missing WT data: (1) WT data are greatly affected by environments. The changeable environment makes it difficult to accurately predict the missing data. (2) Wind farms (WFs) often suffer from data missing problem in large-scale WT's. Fig. 1 shows an example. The wind speed data in 11 out of 16 WT's are missing. Only the data from 5 WT's can be used to do the imputation. (3) Wake flow of WT's leads to a strong nonlinear relationship of the data in different WT's. Thus, simple models cannot achieve expected imputation results in such cases. Therefore, a method that can both handle strong nonlinear relationships and has a strong generation ability to impute the missing data in large-scale WT's is essential for all the data-driven methods in WT fields.

To solve the above problems, a WT data imputation method with multiple optimizations based on generative adversarial networks (GANs) is proposed. GANs [42] are a powerful generative model that have been widely used in many industrial fields. In previous studies [43], the synthetic WT fault data were generated by GANs to train WT fault detection models. In [44], speech synthesis method based on statistical parametric and GANs was put forward and good results were achieved. In [45], a data-driven approach for scenario generation using GANs was proposed and realistic wind and solar data were generated. Data imputation can be essentially considered as a kind of data generation. For this reason, conditional GANs-based deep learning generative model is designed in this paper to impute the missing WT data. Moreover, the imputation model training is improved according to WT data features. The permutation of the training data is optimized and the relationship between the training data and the convolutional kernel (CK) is studied.

The contributions of this paper include:

- (1) A novel deep learning-based generative model is proposed to impute the missing SCADA data of WT's. Benefited from the powerful generative ability of GANs, the proposed method can achieve a good result of imputing the missing data of large-scale WT's.

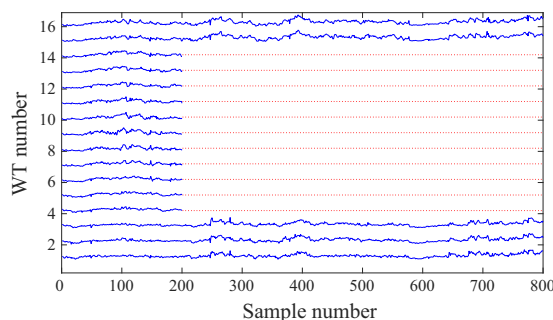


Fig. 1. An example of wind speed data missing case in large-scale WT's.

- (2) The permutation of the training data is optimized so that the convolutional kernel can be better applied. The optimization problem is novelly transformed to a travelling salesman problem (TSP) and two optimization functions are proposed according to WT data features.
- (3) The relationship between the training data and the convolutional kernel is studied in this paper. Restrictions are proposed to utilize the convolutional kernel scientifically so that the imputation effect can be enhanced.

The rest of this paper is organized as follows. Section II introduces the proposed data imputation method. Section III lists the experiments and discussions. The conclusion is summarized in Section IV.

2. The proposed method

2.1. An overview of the proposed method

The proposed method can be divided into three phases, as illustrated in Fig. 2. In the first two phases, the training data are processed by multiple optimizations. By doing so, conditional GANs are introduced to train the generative model in Phase 3. The trained model can be applied to imputing the missing WT data. A brief description of each phase is presented as follows:

Phase 1: CK-based optimization. Restrictions are proposed to optimize the training data according to the severity of data missing cases. Meanwhile, the size of CK can be determined.

Phase 2: WT permutation-based optimization. Object functions are proposed to optimize WT permutation, so that the optimized data can be more suitable for model training.

Phase 3: Data imputation model training. Using the optimized data and the CK obtained in Phase 1, conditional GANs are applied as a supervised learning to train the data imputation model.

To better explain the proposed method, Phase 3 is first described in II.B. The permutation-based optimization (Phase 2) and CK-based optimization (Phase 1) are described in II.C and II.D respectively.

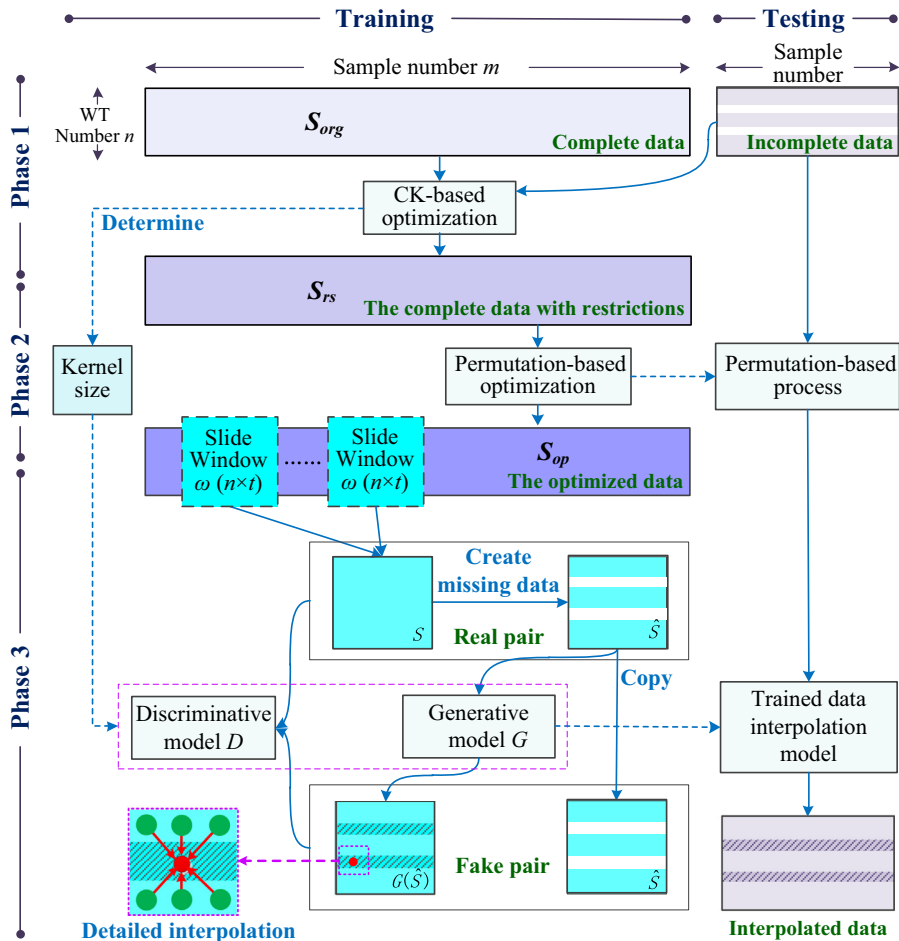


Fig. 2. The data imputation architecture of the proposed method.

2.2. Data imputation based on conditional GANs

In this section, the generative model based on conditional GANs is introduced to impute the missing WT data.

2.2.1. Missing data of WTs and generative adversarial networks

The SCADA data that are collected from a WT can be described as:

$$S_{org} = [(S_{org})_1, (S_{org})_2, \dots, (S_{org})_i, \dots, (S_{org})_n]^T, \quad (S_{org})_i \in \mathbb{R}^{1 \times m} \quad (1)$$

where, S_{org} is one of SCADA data variables (such as wind speed) of all WTs, $(S_{org})_i$ denotes the data of the i th WT, n is the total number of WTs and m is the number of the collected data points. If there is a data missing case, some WT data could not be obtained within a certain period of time. Then, S_{org} can be divided into two parts, the existing data E_p of p WTs and the missing data M_q of q WTs.

$$S_{org} = E_p + M_q, \quad p + q = n \quad (2)$$

Therefore, a data imputation model F_{ip} is used to retrieve the missing data.

$$M_q = F_{ip}(E_p), \quad \frac{q}{p+q} = \varepsilon \quad (3)$$

where, ε is the missing rate. In order to achieve a better imputation result, GANs are employed as the generator to retrieve the missing data. As a powerful generative model, GANs have been effectively used in many fields. Fig. 3 shows the architecture of GANs.

There are two deep learning models in GANs, the generator G and the discriminator D . G and D are trained simultaneously through a minimax game. The purpose of GANs is to enable G to learn the distribution p_G that matches $p_x(x)$ (the distribution of real data). G is trained to generate samples from p_G through processing a noise vector $z [z \sim p_z(z)]$ to a generated sample $G(z)$. Then D is trained to play against G , so that it can distinguish samples from the distribution of real data $p_x(x)$ and the generated data p_G . This minimax game can be expressed as:

$$\min_G \max_D V(D, G) = \min_G \max_D (\mathbb{E}_{x \sim p_x(x)} [\log D(x)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z)))]) \quad (4)$$

2.2.2. Data imputation model based on conditional GANs

In this paper, conditional GANs are applied to training the WT data imputation model.

It is important to properly construct the training data, so that GANs can impute the missing data and their generative capabilities can be fully exerted. In this paper, the training data are constructed according to the structure of GANs. First, the complete data sample set S_{sp} is collected from the optimized data S_{op} (the acquisition of S_{op} is described in section III. C and III.D) by a sliding window $\omega_{n \times t}$, as shown in Fig. 2.

$$S_{sp} = [S_{sp}^{(1)}, S_{sp}^{(2)}, \dots, S_{sp}^{(\tau)}] = [(S_{op}^{(1)} \dots S_{op}^{(t)}), (S_{op}^{(t+1)} \dots S_{op}^{(2t)}), \dots, (S_{op}^{((\tau-1)t+1)} \dots S_{op}^{(\tau t)})], \quad \tau t \leq m \quad (5)$$

where t is the length of $\omega_{n \times t}$. $S_{op}^{(i)}$ is the i th column vector of S_{op} as well as the data of all the n WTs at time i . $S_{op}^{(i)} \in \mathbb{R}^{1 \times n}$. Suppose S is a sample in the sample set S_{sp} , then S can be expressed as a matrix of $n \times t$. Second, according to the severity of the data missing case, the corresponding missing data in certain WTs are removed from S to obtain the incomplete data $\hat{S}$.

$$\hat{S} = [S_1 \times \gamma_1, S_2 \times \gamma_2, \dots, S_n \times \gamma_n]^T \quad (6)$$

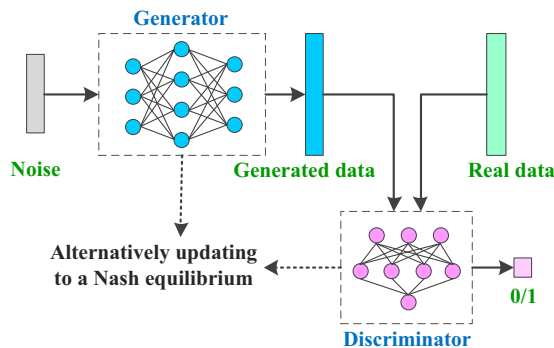


Fig. 3. Architecture of the original GANs.

where S_i is the i th row vector of S . $\gamma_i = 1$ if the data of i th WT exists, otherwise $\gamma_i = 0$. Then, S and $\hat{S}$ are combined as the real data pair, as shown in Fig. 2.

Using the well-constructed data, the data imputation model can be trained. In this step, convolutional neural networks are used in both G and D . The deep learning networks can effectively handle the nonlinear relationships. In this way, one single missing data point can be imputed by the information from all its surrounding data points (including the historical data, the current data and the future data), as red arrows marked in Fig. 2. As a result, a better imputation effect can be achieved.

In the training process, $\hat{S}$ is processed by generator G and the generated data $G(\hat{S})$ is acquired. Within $G(\hat{S})$, the missing data are imputed by G through data generation. However, the imputation may not achieve expected performance at first. Consequently, $G(\hat{S})$ and $\hat{S}$ are combined as the fake data pair, as shown in Fig. 2. Unlike the original GANs, conditional GANs use data pairs to train discriminator D . The real data pair $[S, \hat{S}]$ and the fake data pair $[G(\hat{S}), \hat{S}]$ are combined to be processed by discriminator D . By doing so, D is trained to distinguish real data pairs from fake ones gradually. Therefore, the loss function of the generator G can be expressed as (7).

$$L_{ip}^G(G) = -\mathbb{E}_{S \sim p_S(S), \hat{S} \sim p_{\hat{S}}(\hat{S})} [\log(D(G(\hat{S}), \hat{S}))] + \lambda \|G(\hat{S}) - S\|_1 \quad (7)$$

There are two terms in the loss function: the former tries to cheat D by updating the parameters of G and the latter is designed to make the generated data more approximate to the real data. Accordingly, the loss function of discriminator D can be expressed as (8).

$$L_{ip}^D(D) = -\mathbb{E}_{S \sim p_S(S), \hat{S} \sim p_{\hat{S}}(\hat{S})} [\log D(S, \hat{S})] - \mathbb{E}_{S \sim p_S(S), \hat{S} \sim p_{\hat{S}}(\hat{S})} [\log(1 - D(G(\hat{S}), \hat{S}))] \quad (8)$$

There are also two terms in the loss function: the former is to detect the real data pair and the latter is to detect the fake data pair. It can be found that the complete data and the incomplete data are trained together both in real data pairs and in fake ones. Therefore, the generator G can generate the complete data that match the real data.

As the training process continues, D will gain a stronger ability to distinguish the real data pair from the fake one, whereas G can gradually generate the data more similar with the real data. Consequently, the well-trained G can be applied in WT data imputation.

Finally, the trained model can be used for testing. The trained generator G is used as the model of data imputation. Unlike in the training phase, the missing data sample, rather than the data pair, is processed by G in model testing. The output of G is the imputed data.

2.3. Permutation optimization of WT data imputation

The permutation of WTs is a key factor in WT data imputation. The training data need more two-dimensional features (both rows and columns in the data matrix should be smooth and highly-correlated) to match CK used in the imputation model, as illustrated in Fig. 4. Therefore, an optimized WT permutation of the training data is very important for data imputation. In order to get a better optimization effect, object functions are proposed in this paper. First, the correlation between the data collected from two WTs are calculated by Pearson correlation coefficient (PCC), as shown in (9).

$$pcc(x, y) = \frac{E(xy) - E(x)E(y)}{\sqrt{E(x^2) - E^2(x)} \sqrt{E(y^2) - E^2(y)}} \quad (9)$$

Then, the optimization object functions are designed to provide the training data with more two-dimensional features:

(1) High correlations. If the data in any two adjacent WTs are highly-correlated within a permutation, the training data would have good two-dimensional features. Therefore, the first object function of WT permutation is proposed as (10).

$$O_1 = -\max_p \frac{\sum_{i=1}^{n-1} pcc(P(S_{org})_i, P(S_{org})_{i+1})}{n-1} \quad (10)$$

where P is the operation of adjusting WT permutation. S_{org} is the original data and $P(S_{org})$ is the adjusted data. $P(S_{org})_i$ and $P(S_{org})_{i+1}$ are the data of two adjacent WTs in $P(S_{org})$.

(2) Smoothness. The deviation value of correlations of the data in any two adjacent WTs in a permutation should not be too high, so that the training data can be smooth. Accordingly, the second object function is proposed as (11).

$$O_2 = -\max_p \left(\min \left(pcc(P(S_{org})_j, P(S_{org})_{j+1}) \right) \right), \quad j = 1, 2, \dots, n-1 \quad (11)$$

The final optimization objective function can be expressed as (12).

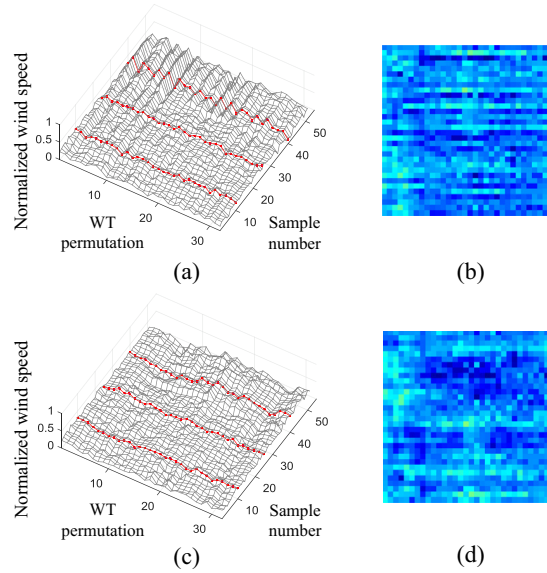


Fig. 4. Different two-dimensional features of training data caused by different WT permutations. (a) A weak permutation (with few two-dimensional features). (c) A strong permutation (with more two-dimensional features). (b) and (d) are the imaged data of (a) and (c).

$$O^* = O_1 + \lambda O_2 = -\max_p \left(\frac{\sum_{i=1}^{n-1} pcc(P(S_{org})_i, P(S_{org})_{i+1})}{n-1} + \lambda \min(pcc(P(S_{org})_j, P(S_{org})_{j+1})) \right), \quad j = 1, 2, \dots, n-1 \quad (12)$$

The optimization of WTs permutation has the following features: (1) each WT can only be used once and (2) the correlation (or weight) of any two WTs can be calculated. Therefore, this optimization problem can be properly regarded as a TSP. In this paper, genetic algorithm (GA) is applied to realizing the optimization.

Inspired by the representations and operators in [46], the GA used in this paper adopts order-based crossover and the displacement mutation, which are more suitable for the optimization of permutation.

2.4. Training data restrictions based on convolutional kernel

Another important factor of the proposed method is the training data in CK. Fig. 5 shows an example. When there are data missing cases in large-scale WTs, there should be enough information in each CK traversing in a training sample to obtain good imputation results. Therefore, restrictions are proposed and added in the permutation of the training data.

(1) There should be at least one existing WT data in every CK traversal of the training sample. Therefore, the first restriction is proposed as (13).

$$\forall K_\delta [P(S_{org})] \geq 1 \quad (13)$$

where, K_δ denotes the CK and δ is the size of the CK. $K_\delta[\cdot]$ is the number of data-containing WTs in CK.

(2) The number of the existing data in every CK traversal should be averaged to ensure that there is the same amount of information in each CK to achieve a balanced imputation effect. The second restriction is proposed as (14).

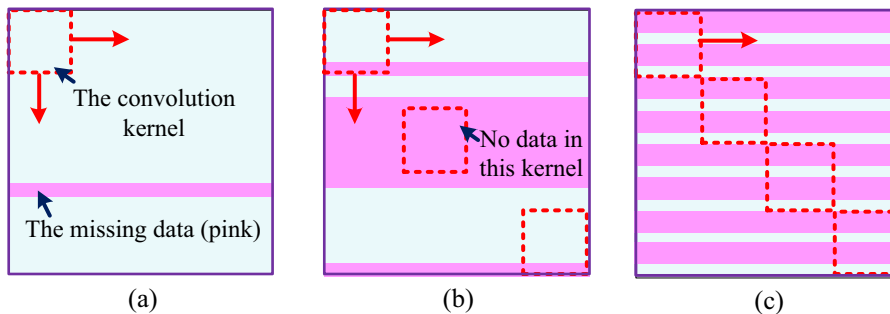


Fig. 5. Three data missing cases and the CK. (a) There are data in CK in each traversal. (b) There are no data in CK in some traversals. (c) For each traversal, there are always two WT data in CK.

$$\text{dev}(K_\delta[P(S_{org})]_1, K_\delta[P(S_{org})]_2, \dots, K_\delta[P(S_{org})]_r) \leq \varphi \quad (14)$$

where, dev denotes the computation of standard deviation, $K_\delta[P(S_{org})]_i$ is the number of data-containing WTs of the i th CK traversal, r is the total traversal number and φ is the threshold.

Based on the optimization of WT permutation and the restrictions according to CK, the data can be more suitable for conditional GANs-based model training and the data imputation can be more effective. The complete algorithm of WT missing data imputation is shown in Algorithm 1.

Algorithm 1: WT Data Imputation Model Training

Input: S_{org} – the complete data, $S_{org} \in \mathbb{R}^{n \times m}$, n – the number of WTs, m – the number of data points, k – the size of CK, η – learning rate.

Output: G – data imputation model.

- 1: Adjust k and WT permutation according to restrictions:
 - 2: $(S_{rs}, k) \leftarrow \text{CK_optimization}(S_{org})$,
 - 3: refer to Eq. (13) and (14).
 - 4: Optimize the WT permutation of the training data:
 - 5: $S_{op} \leftarrow \text{PM_optimization}(S_{rs})$, refer to Eq. (12).
 - 6: Make the incomplete data M :
 - 7: $M_{op} \leftarrow \text{extracting}(S_{op})$, refer to Eq. (6).
 - 8: Construct samples for model training:
 - 9: $(S_{sp}, M_{sp}) \leftarrow \omega_{(n \times n)}(S_{op}, M_{op})$, refer to Eq. (5).
 - 10: **for** number of training iterations **do**
 - 11: **for all** training sample $(S, \hat{S})$ in (S_{sp}, M_{sp}) **do**
 - 12: imputing the missing data $\hat{S}$ into $\hat{S}$:
 - 13: $(\hat{S}, \hat{S}) \leftarrow G(S, \hat{S})$.
 - 14: update θ_D with fixed θ_G :
 - 15: $\theta_D \leftarrow \theta_D - \eta \nabla_{\theta_D} L_{imp}^D[(S, \hat{S}), (\hat{S}, \hat{S})]$.
 - 16: update θ_G with fixed θ_D :
 - 17: $\theta_G \leftarrow \theta_G - \eta \nabla_{\theta_G} L_{imp}^G[(S, \hat{S}), (\hat{S}, \hat{S})]$.
 - 18: **end for**
 - 19: **end for**
-

2.5. Evaluation indexes of data imputation

In order to examine the performance of data imputation methods, the following evaluation indexes are introduced in this paper. The following three indexes are selected to evaluate the data imputation for a single WT: mean absolute error (MAE) in [47], mean square error (MSE) in [28] with adjustment and normalized root mean square error (NRMSE) in [34]. Expressions of the three indexes are shown in (15), (16) and (17).

$$MAE = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t| \quad (15)$$

$$MSE = \frac{1}{n} \left(\sum_{t=1}^n (y_t - \hat{y}_t)^2 \right)^{\frac{1}{2}} \quad (16)$$

$$NRMSE = \frac{\left(\sum_{t=1}^n (y_t - \hat{y}_t)^2 \right)^{\frac{1}{2}}}{\sum_{i=1}^n y_t} \quad (17)$$

Moreover, the averaged value and the standard deviation of MAE, MSE and NRMSE are used to evaluate the data imputation for multiple WTs.

3. Experiments and discussions

3.1. Experiment settings

To verify the effectiveness of the method presented in this paper, several experiments are carried out. A wind farm in northern China is selected for experiments. Fig. 6 shows the pictures of the wind farm and the SCADA system. 32 WTs



Fig. 6. The wind farm and the SCADA system used in experiments. (a) The wind farm. (b) The SCADA system.

are studied in this wind farm and the SCADA data with a sampling interval of 10 min are collected. In order to achieve better comparison results, the complete data are used in all the experiments. The missing data are acquired by removing the specified data from the complete data.

The parameters of the proposed method are set as follows: convolutional kernel of GANs is initialized as 8×8 , the size of the slide window is set as 32×32 , the initial population size of GA is set as 100, and the maximum generation number is set as 200. The SCADA data of altogether 300 days are used in experiments. The data of 240 days are used for training and the rest data of 60 days are used for testing. These SCADA data are exported from the SCADA system in the wind farm and they occupy a disc space of 672 MB.

The experiments in this section are conducted using a workstation with the following configuration: Intel(R) Xeon(R) CPU E5-2640 v4 @ 2.40 GHz, 32G Memory, NVIDIA GeForce GTX 1080 Ti GPU and 1 TB hard disc. When the data in 24 out of 32 WTs are missing, the computation cost of the proposed method is about 4885s in training (with the data of 240 days) and 3.48s in testing (with the data of 60 days).

In the following experiments, the variable “wind speed” is chosen as the imputed data. Only the data between the cut-in and the cut-off speed are used. Data imputation is adopted with an aim of retrieving WT missing data with the existing data collected from other WTs.

The rest of this section is organized as follows. WT data imputation experiments are conducted in Section III. B to test the effectiveness of the method in imputing the data missing in small-scale WTs, and in III.C to test the effectiveness of the method in imputing the missing data in large-scale WTs. The proposed permutation optimization and CK-based training data restrictions are verified respectively in Section III.D and Section III.E.

3.2. Data imputation for the data missing in small-scale WTs

The small-scale data missing cases (data missing in several WTs) happens frequently in wind farms. In this subsection, two groups of experiments (data imputation for one WT and four WTs) are conducted to verify the effectiveness of the proposed method.

3.2.1. Data imputation for one WT

In this experiment, the wind speed data in only one WT (WT29) is missing. The data of the other 31 WTs are used to realize data imputation.

In order to verify the effectiveness of the proposed GANs-based data imputation method (ipGAN), three commonly-used methods [ARMA method, back propagation neural network (BPNN) method, and SVR method] and three deep learning methods [deep belief network (DBN) method, convolutional neural network (CNN) method, and de-noised autoencoder (DAE) method] are used to compare with the proposed method. The process and parameters of these methods are listed as follows: (1) ARMA method. ARMA regression model is trained by the input time series vectors integrated by the existing data. Its parameters, order of A polynomial, B polynomial and C polynomial, are respectively set as 2, $[1, 1, \dots, 1]$ (the length is the number of WTs with data) and 2. (2) BPNN method. A three-layered regression model is trained to map the existing data to the missing data. The epoch number and learning rate are set as 100 and 0.00001. (3) SVR method. A Gaussian-kernel SVR regression model is trained to predict the missing data with the existing data. Epsilon and Penalty factor of the model are set as 0.1 and 1.0, respectively. (4) DBN method. A deep belief network is used to pre-train the initial values of the weights of a forward neural network and this network is used to impute the missing data. (5) CNN method. A six-layered convolutional neural network is trained as a regression model for data imputation. (6) DAE method. A de-noised autoencoder is trained to initialize a feed forward neural network to impute the missing data.

The experiment results are shown in Fig. 7 and the values of evaluation indexes are listed in Table 1.

As can be seen in Fig. 7, the proposed method has the best performance. The data imputed by the proposed method are more approximate to the real data. Moreover, the evaluation indexes in Table 1 show that the proposed method has a better

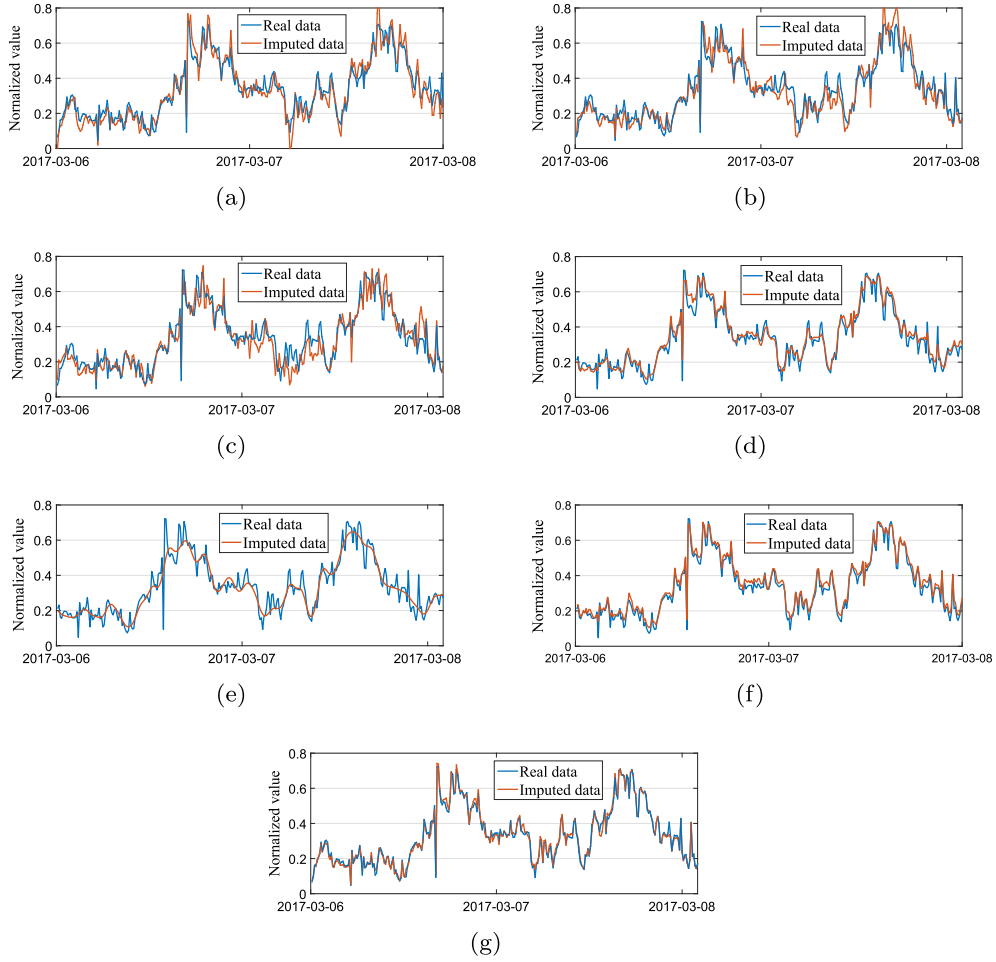


Fig. 7. Imputation results of the wind speed data of WT 29. (a) ARMA. (b) BPNN. (c) SVR. (d) DBN. (e) CNN. (f) DAE. (g) ipGAN.

Table 1
Data Imputation Results of WT 29.

Method	MAE	MSE	NRMSE
ARMA	0.0377	0.0028	0.0084
BPNN	0.0379	0.0029	0.0086
SVR	0.0422	0.0033	0.0100
DBN	0.0267	0.0020	0.0059
CNN	0.0374	0.0031	0.0093
DAE	0.0233	0.0016	0.0046
ipGAN	0.0112	0.0009	0.0027

imputation effect, which may attribute to the outstanding generating ability of the deep learning model trained by GANs. The experiment results indicate that the proposed method is effective in imputing data for one WT.

3.2.2. Data imputation of four WTs

In order to further verify the effectiveness of the proposed method for the data imputation in small-scale WTs, another group of experiments have been conducted. In these experiments, the data of four WTs (WT 01, WT 09, WT 17 and WT 31) are removed and the data of the remaining 28 WTs are used to impute the missing data. The experiment results are shown in Fig. 8.

It can be seen from the three evaluation indexes, the proposed method has the best imputation performance. For the data imputation of WT 01, WT 09, WT 17 and WT 31, the performance of the proposed method is much better than those of the other six methods. Therefore, it can be concluded from the experiment results that the proposed method is effective for the data imputation in small-scale WTs.

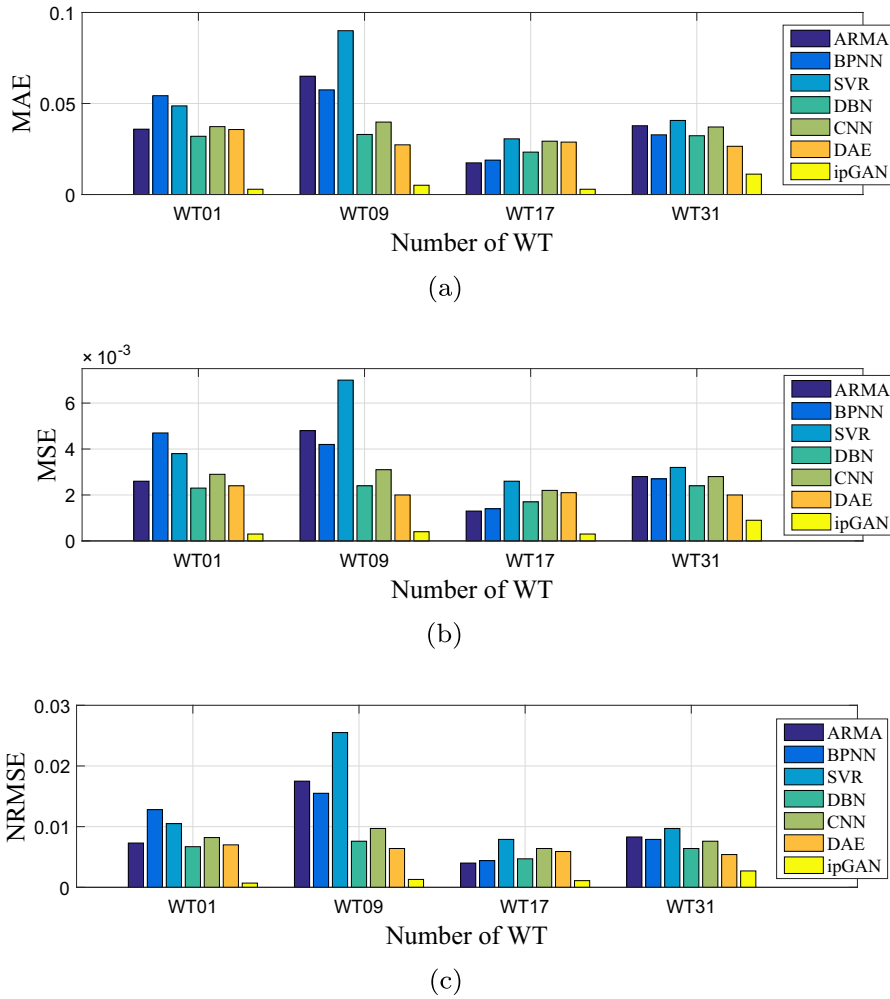


Fig. 8. Imputation result of wind speeds of WT 01, WT 09, WT 17 and WT 31. (a) MAE. (b) MSE. (c) NRMSE.

3.3. Data imputation for the data missing in large-scale WTs

Sometimes, wind farms suffer from data missing in large-scale WTs. In such case, more than half of the data could be lost. In this subsection, experiments are conducted to verify the effectiveness of the proposed method to impute the missing data in large-scale WTs.

3.3.1. Effectiveness of imputing the missing data in large-scale WTs

Two groups of experiments are conducted in this subsection: (1) the data of 16 WTs are missing and the data of the other 16 WTs are used to impute the missing data; (2) the data of 24 WTs are missing and the data of the other 8 WTs are used to do the data imputation. Table 2 shows the evaluation results of the experiments. Fig. 9 and Fig. 10 show the imputation details of WT 07 and WT 23.

As the results indicate, (1) the proposed method has the best averaged performance in the two groups of experiments (for 16 WTs and for 24 WTs). (2) Among the seven methods, the proposed method has relatively low standard deviations of the three evaluation indexes, which means that the proposed method has a balanced data imputation effect of all the WTs with the missing data. Moreover, the data curves in Fig. 9 and Fig. 10 show that the data imputed by the proposed method fit the real data better than those of the other methods.

3.3.2. Imputation performance influenced by the number of data-missing WTs

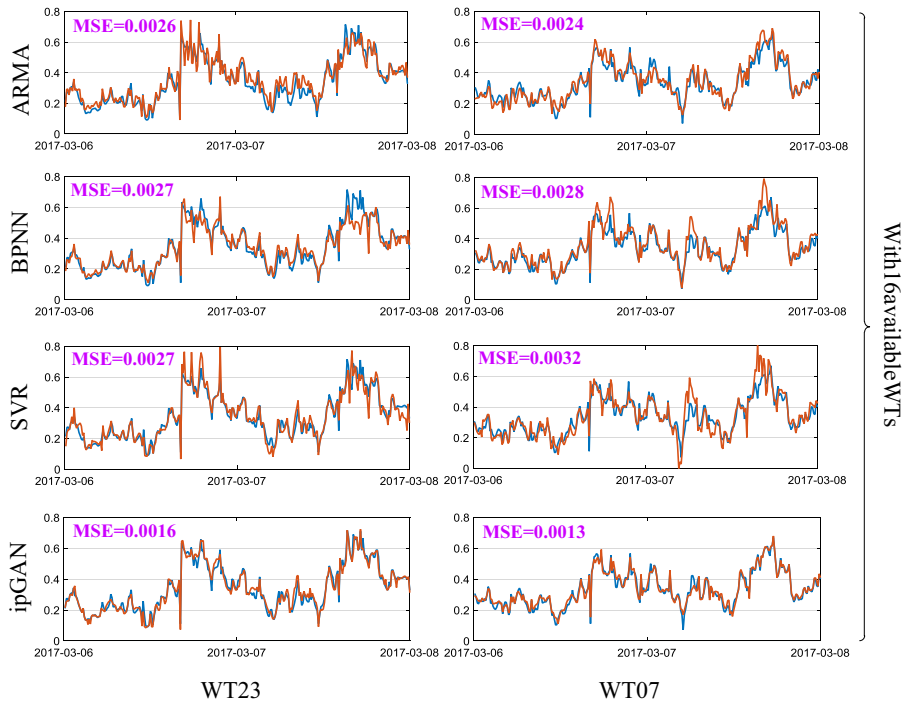
In order to further verify the effectiveness of the proposed method, the relationship between the number of data-missing WTs and the imputation performance is studied in this part. Four groups of experiments (the number of WTs with data is 28, 24, 16 and 4 respectively) are carried out. Fig. 11 shows the performance curves of different methods.

Table 2

Data imputation results of 16 WTs and 24 WTs.

WTs	Method	MAE				MSE				NRMSE			
		Max.	Min.	Ave.	Sd.	Max.	Min.	Ave.	Sd.	Max.	Min.	Ave.	Sd.
16 WTs	ARMA	0.1023	0.0186	0.0414	0.0216	0.0072	0.0014	0.0031	0.0015	0.0244	0.0036	0.0095	0.0054
	BPNN	0.0651	0.0209	0.0434	0.0132	0.0052	0.0016	0.0034	0.0011	0.0173	0.0050	0.0104	0.0038
	SVR	0.0894	0.0316	0.0580	0.0118	0.0076	0.0027	0.0048	0.0016	0.0271	0.0068	0.0149	0.0060
	DBN	0.0620	0.0133	0.0362	0.0106	0.0043	0.0009	0.0027	0.0010	0.0132	0.0027	0.0079	0.0035
	CNN	0.0583	0.0193	0.0392	0.0101	0.0041	0.0014	0.0030	0.0009	0.0131	0.0037	0.0090	0.0036
	DAE	0.0569	0.0142	0.0359	0.0109	0.0036	0.0011	0.0024	0.0010	0.0122	0.0031	0.0071	0.0034
	ipGAN	0.0489	0.0019	0.0211	0.0099	0.0035	0.0002	0.0017	0.0011	0.0119	0.0005	0.0051	0.0032
8 WTs	ARMA	0.1094	0.0276	0.0498	0.0172	0.0077	0.0021	0.0037	0.0012	0.0257	0.0053	0.0113	0.0043
	BPNN	0.0952	0.0292	0.0547	0.0161	0.0084	0.0022	0.0043	0.0015	0.0268	0.0065	0.0132	0.0051
	SVR	0.0780	0.0316	0.0498	0.0117	0.0062	0.0024	0.0039	0.0009	0.0200	0.0068	0.0120	0.0034
	DBN	0.0648	0.0258	0.0414	0.0109	0.0051	0.0015	0.0031	0.0010	0.0138	0.0048	0.0093	0.0033
	CNN	0.0608	0.0279	0.0426	0.0108	0.0048	0.0016	0.0033	0.0009	0.0141	0.0051	0.0099	0.0034
	DAE	0.0710	0.0263	0.0401	0.0102	0.0049	0.0012	0.0029	0.0011	0.0149	0.0042	0.0087	0.0033
	ipGAN	0.0592	0.0026	0.0333	0.0104	0.0043	0.0003	0.0025	0.0009	0.0132	0.0010	0.0076	0.0032

Ave. denotes average and Sd. denotes standard deviation.

**Fig. 9.** Data imputation results of WT 07 and WT 23 when there are data missing cases in 16 WTs out of 32 WTs. The ARMA method, BPNN method and SVR method are used to compare with the proposed method.

The experiment results show that: (1) as the number of WTs with data (N-WD) decreases, the imputation effect declines using most of the methods. (2) The proposed method has the lowest value of the evaluation indexes among the seven methods and has the best performance in each experiment.

Therefore, it can be concluded that: (1) the proposed method is effective in dealing with the data missing problem in large-scale WTs and (2) the proposed method has good performance in different data missing cases.

3.4. Verification of the proposed permutation optimization

In this subsection, two groups of experiments are carried out to verify the proposed training data permutation optimization. The same training data with different WT permutations are used in the experiments. Permutation 1 employs the sequential permutation of WT number (WT 01, WT 02, ..., WT 32), whereas Permutation 2 uses the optimized permutation. Fig. 12 illustrates the training data of the two permutations in images and Table 3 lists the values of evaluation indexes.

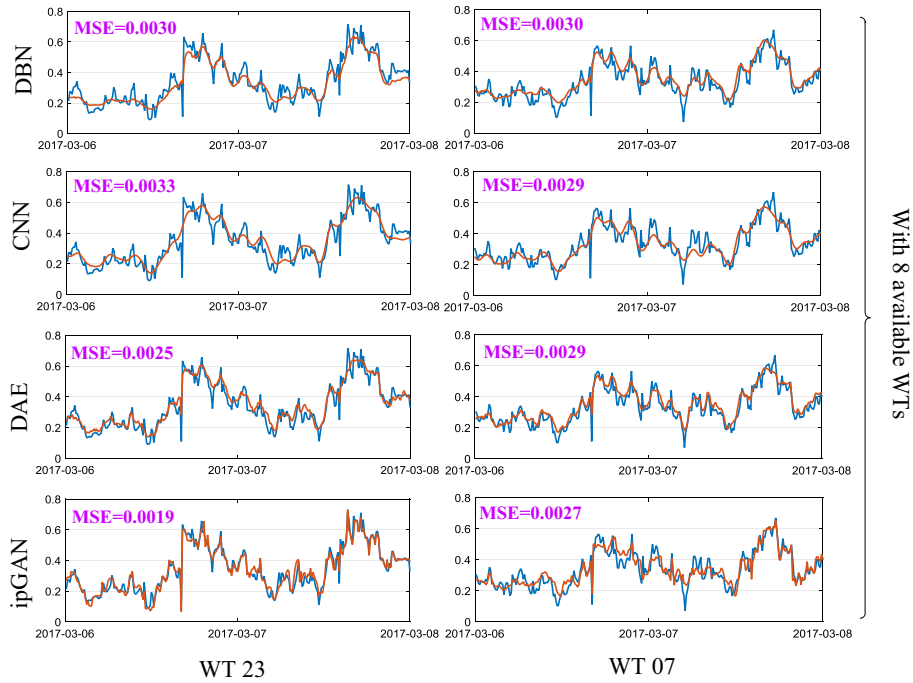


Fig. 10. Data imputation results of WT 07 and WT 23 when there are data missing cases in 24 WTs out of 32 WTs. The DBN method, CNN method and DAE method are used to compare with the proposed method.

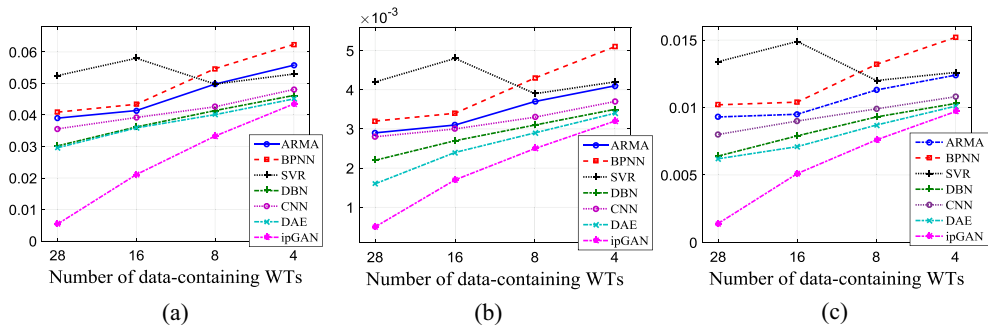


Fig. 11. Data imputation performances influenced by the number of data-containing WTs (the total number of WTs is 32). (a) MAE. (b) MSE. (c) NRMSE.

It can be found from Fig. 12 that compared with the sequential data, the optimized data have more two-dimensional features. Therefore, the CK-based deep learning method can be more effectively applied to imputing the missing data. The evaluation index values in Table 3 show that the imputation effect after the training data optimization is improved by about 15% than that before the optimization. Moreover, in testing phase, the operation of the optimization (adjusting data permutation) takes very little time. The time cost of the proposed method with and without the optimization is 3.5632s and 3.5630s respectively. The experiment results indicate that the permutation optimization can enhance the imputation effect of the proposed method.

3.5. Verification of the proposed training data restrictions based on CK

In this subsection, four groups of experiments are carried out to verify the effectiveness of the proposed training data restrictions based on CK. Four types of training data with different permutations are designed in Fig. 13. In order to make a better comparison, the size of CK is fixed. Through data traversing, CK will cover different numbers of data-containing WTs. (1) Type 1: the WT number covered by CK (NCK) is uneven, with a range of 0–4. (2) Type 2: The range of NCK is 1–3. (3) Type 3: NCK is 2, but the space between two adjacent data is not equal. (4) Type 4: NCK is 2 and the space between two adjacent data is equal. The experiment results are listed in Table 4.

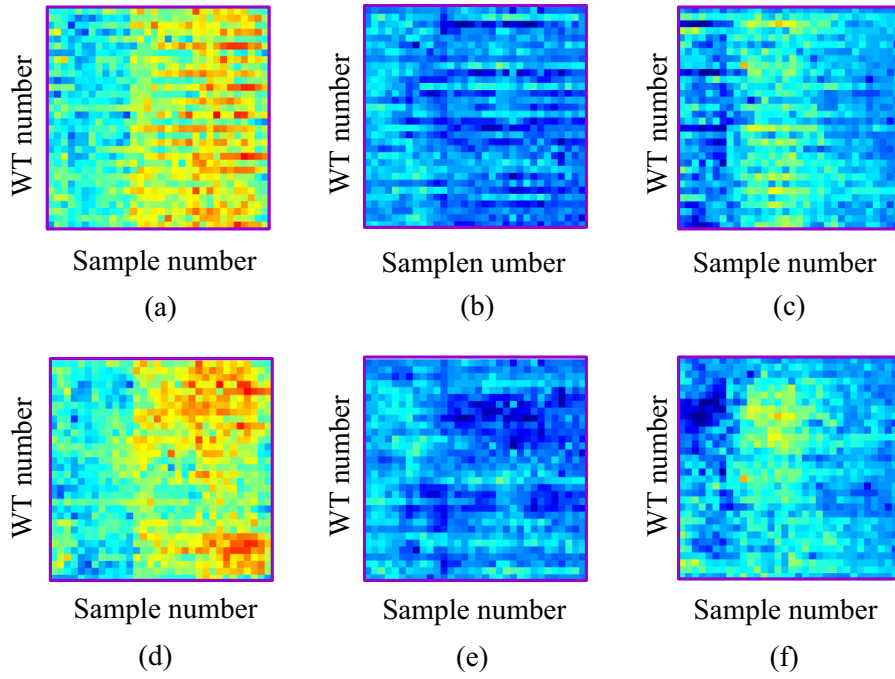


Fig. 12. Illustration of the imaged data with different WT permutations. There are three data pairs (a)–(d), (b)–(e) and (c)–(f): (a), (b) and (c) are the imaged data with the sequential permutation, while (d), (e) and (f) are the imaged data with the optimized permutation.

Table 3

Comparison of the Imputation Results with and without Permutation Optimization.

Evaluation Index	ipGAN before optimization	ipGAN after optimization
Ave. of MAE	0.0521	0.0435
Sd. of MAE	0.0186	0.0154
Ave. of MSE	0.0037	0.0032
Sd. of MSE	0.0013	0.0011
Ave. of NRMSE	0.0110	0.0097
Sd. of NRMSE	0.0039	0.0036

Ave. denotes average and Sd. denotes standard deviation.

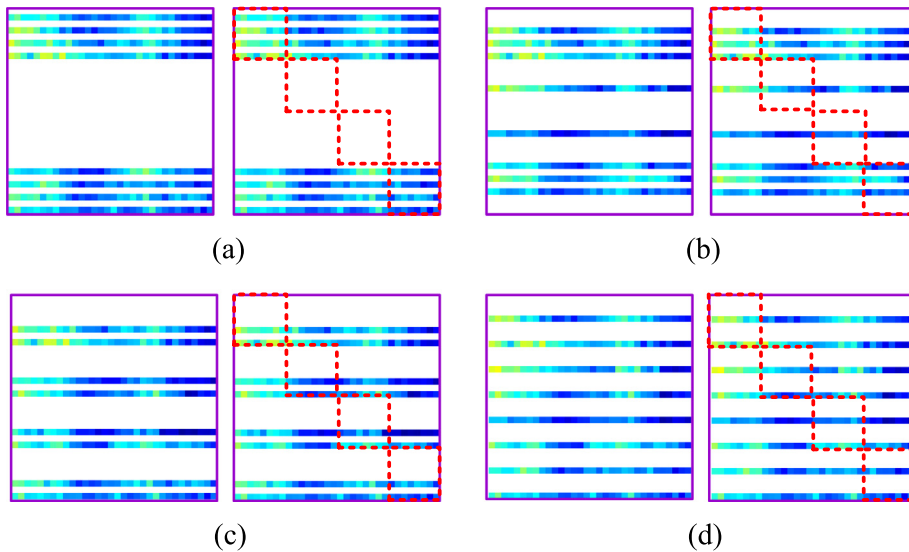


Fig. 13. Different training data styles in experiments. (a) Data type 1. (b) Data type 2. (c) Data type 3. (d) The data type optimized by the proposed method.

Table 4
Experiment Results With Different Styles of Training Data.

Data type	MAE		MSE		NRMSE	
	Ave.	Sd.	Ave.	Sd.	Ave.	Sd.
Data type 1 shown in Fig. 13 (a)	0.0459	0.0191	0.0034	0.0013	0.0104	0.0039
Data type 2 shown in Fig. 13 (b)	0.0350	0.0157	0.0028	0.0011	0.0082	0.0036
Data type 3 shown in Fig. 13 (c)	0.0348	0.0151	0.0027	0.0012	0.0081	0.0038
The optimized data type shown in Fig. 13 (d)	0.0333	0.0104	0.0025	0.0009	0.0076	0.0032

Ave. denotes average and Sd. denotes standard deviation.

Experiment results in Table 4 indicate that the performance of the method with the optimized data type has been improved by about 20% on average compared with that of the method with the other data types. Maybe it is because CK processes the same amount of data in the same permutation in each traversal of the proposed data type. Other data types may have a better effect in some cases, but a poor performance on the whole. The time costs of the proposed method using the four different data types are 3.4848 s, 3.4846 s, 3.4846 s and 3.4847 s, respectively. Therefore, it can be concluded that the proposed restrictions can enhance the effect of the proposed method.

3.6. Discussions

The experiment results in this section show that the proposed method is effective in imputing the missing data of large-scale WTs. The imputation model of the proposed method is built offline. First, the optimization procedures are conducted offline using historical data. Once the optimized permutation is obtained, the training data are adjusted to the optimized data in accordance with the optimized permutation. Then, the GANs-based imputation model is trained with the optimized data. Using the optimized permutation and the trained model, the missing data can be imputed.

The proposed method is capable of dealing with the amount of SCADA data in a wind farm. With limited number of WTs in a wind farm, the amount of SCADA data could not be too large. Therefore, the influence of the sample number to the proposed method is limited.

The proposed method aims to impute the missing data in their original form. Therefore, noises, which are a part of the original data, are not removed through the imputation process. The influence of the noise on the proposed method is limited, as the imputation model is trained with the original noise-containing data. Moreover, the proposed method is also beneficial for data cleaning. Data cleaning is an important technology for improving data quality and it has achieved remarkable results in recent years [48,49]. We will continue to study data cleaning in the future.

Moreover, WTs are installed in different places in a wind farm. Influenced by wake effects and the terrain [50,51], on which these WTs are installed, data of different WTs are inconsistent. Consequently, it is difficult to realize the data imputation by replacing the missing data with the data from other WTs. Therefore, the data of all the other WTs are used to impute the missing data in the proposed method. In this paper, the missing data are acquired by removing the specified data from the complete data, so that the supervising data are correct and they can properly supervise the model training. This method of acquiring missing data is easy to conduct and it can create various data missing cases.

4. Conclusion

This paper proposes a novel WT data imputation method with multiple optimizations based on GANs to solve the problem of data missing in large-scale WTs. First, the conditional GANs are introduced to impute the missing data of large-scale WTs through data generation. Then, according to WT training data features, the permutation of WTs is optimized. Two object functions are proposed and the optimization problem is creatively transformed to a TSP. The relationship between the training data and CK is also studied. Restrictions are proposed to make the training data more adaptable for CK. Experiments are carried out to verify the method using real WT data. The results verify that the proposed method is effective.

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